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Part 1: Concept

Explain the difference between classification, detection, and segmentation. Give one workplace example of each.

Image classification is the ability of a model to identify what is in an image. Object detection is the ability of a model to detect and classify objects in an image and identify their locations using bounding boxes. Image segmentation is the ability of a model to identify the outline of an object by determining which pixels belong to it.

Image classification can be used in online shopping websites to classify products into different categories, such as shoes, shirts, and pants. When customers want to sort products by type, the AI needs to know which category each product belongs to so it can help customers find what they are looking for.

Object detection can be used in airport security to identify different objects inside passengers' luggage during security screening. The AI can detect objects and show their locations using bounding boxes. This could help security workers identify suspicious items more efficiently.

Image segmentation can be used in hospitals to help doctors analyze tumors in medical images. Doctors need to understand the shape and size of a tumor to help determine how serious it might be. Image segmentation can show the outline of the tumor and help doctors understand how much area it covers.

What is Non-Maximum Suppression (NMS), and why does YOLO use it?

Non-Maximum Suppression (NMS) is a technique that removes unnecessary overlapping bounding boxes around the same object. YOLO uses NMS to keep the bounding box with the highest confidence score and remove duplicate boxes. Without NMS, the model might detect the same object multiple times, making the results confusing and less accurate.

What does Intersection over Union (IoU) measure, and what does a high IoU score mean?

Intersection over Union (IoU) measures how much the predicted bounding box overlaps with the actual bounding box of an object. A high IoU score means the model is more accurate at identifying the location of the object. A lower IoU score means the predicted bounding box does not match the actual location as closely.

Part 2: Hands-on experience

When you ran YOLO11 on your own image, did it detect everything you expected? Did it detect anything that was wrong? Why do you think it succeeded or failed?

When I tested YOLO11 on my image, the model detected all five people that I expected. It also detected a laptop, which I did not expect the model to recognize. One of the person detections had a low confidence score of 28%, but the model still correctly identified the person. Overall, I think the model performed well because it was able to identify all five people and the laptop without missing any of them.

When you compare detection and segmentation side by side, name one situation where the box would be enough and one situation where you would really need the mask.

A bounding box would be enough for airport security because security workers mainly need to know the approximate location of a suspicious object inside luggage. They can open the luggage and inspect the object more closely. However, doctors need more detailed information about a tumor because they cannot easily examine it directly. Image segmentation can help doctors identify the tumor's location, size, and shape. This information can help them understand the condition and plan for surgery if needed.

SAM 2 does not output category labels. Why is that an important difference from YOLO?

SAM 2 can provide a clear outline of an object, but it does not automatically identify what the object is. This is important because we might need to know what the object is, not just its shape. YOLO can help by identifying the object and providing a bounding box. SAM 2 can then use the bounding box to create a more detailed outline of the object. By using both models together, we can identify the object and understand its shape more accurately.

Part 3: Trade-offs

Confidence threshold trade-off. Give one application where you would want a low threshold (more sensitive) and one application where you would want a high threshold. Explain why.

Changing the confidence threshold affects how many objects YOLO detects. A lower confidence threshold allows the model to detect more objects, but it could also increase the number of false alarms. A higher confidence threshold reduces the number of detections, but it could cause the model to miss some objects.

For airport security, I would choose a threshold somewhere in between because security workers cannot ignore suspicious objects just because the AI model has low confidence. However, I also would not want too many false alarms because it could waste time checking luggage that does not contain anything suspicious. I think finding a balance between the two is important, while making sure potential threats are not overlooked.

Precision versus recall. For a workplace safety system that detects unsafe behavior on a factory floor, do you want higher precision or higher recall? What about for a system that tags products in a customer's photo upload?

For a workplace safety detector, I think recall is more important than precision because missing a worker who is not wearing a helmet could be dangerous. I would rather have the AI give a false alarm than miss a real safety violation. A supervisor could check the alerts to determine whether they are real or not. This could help prevent accidents and protect workers.

For a customer photo product tagger, I think precision is more important because I want the AI to correctly identify and label the products. I would rather have the AI assign fewer tags that are more accurate than assign more tags that could be incorrect. Incorrect tags could confuse customers and make it harder for them to find the products they are looking for.

When would you reach for a specialist model like YOLO11-seg, and when would you reach for a foundation model like SAM 2?

YOLO11-seg is a specialist segmentation model that can identify objects and create segmentation masks. One of its advantages is that it is fast and can identify different types of objects. I think YOLO11-seg would be better suited for real-time security cameras because it needs to detect people quickly. However, one of its limitations is that it mainly recognizes objects from the categories it was trained on.

SAM 2 is a foundation segmentation model that can create detailed outlines of many different objects using prompts. One of its advantages is that it can segment objects even if it does not know what they are. For example, in medical imaging, doctors might need to understand the size and shape of a suspected tumor before determining what it is. SAM 2 could help outline the area for further examination. However, SAM 2 does not automatically identify object categories like YOLO11-seg, and it generally requires more computing resources.

I think both models are useful, but which one is better depends on what we are trying to accomplish.